

• 1. **Introduction** to the course and its objectives.

• 2. **Mathematical Preliminaries** including sets, functions, and limits.

• 3. **Calculus I** covering differentiation and integration techniques.

• 4. **Calculus II** focusing on advanced integration and applications.

• 5. **Linear Algebra** introducing vectors, matrices, and linear transformations.

• 6. **Differential Equations** exploring ordinary and partial differential equations.

• 7. **Probability and Statistics** covering basic concepts and distributions.

• 8. **Complex Analysis** introducing functions of complex variables.

• 9. **Topology** exploring the properties of spaces and continuity.

• 10. **Final Review** and preparation for the final examination.

• 11. **Advanced Topics** including a brief overview of modern mathematical research.

• 12. **Guest Lectures** by experts in various fields of mathematics.

• 13. **Research Projects** where students work on open problems.

• 14. **Mathematical Writing** workshops to improve communication skills.

• 15. **Mathematical Modeling** applications in science and engineering.

• 16. **Mathematical History** and the evolution of mathematical thought.

• 17. **Mathematical Philosophy** and the foundations of mathematics.

• 18. **Mathematical Computing** using software tools for calculations.

• 19. **Mathematical Education** and the role of mathematics in society.

• 20. **Mathematical Art** and the intersection of mathematics and aesthetics.

• 21. **Mathematical Games** and recreational mathematics.

• 22. **Mathematical Puzzles** and problem-solving techniques.

• 23. **Mathematical Olympiads** and competitive problem solving.

• 24. **Mathematical Literature** and the history of mathematical writing.

• 25. **Mathematical Software** and the use of computers in mathematics.

• 26. **Mathematical Networks** and the structure of mathematical research.

• 27. **Mathematical Communities** and the role of collaboration.

• 28. **Mathematical Careers** and the path to a research position.

• 29. **Mathematical Ethics** and the responsibility of mathematicians.

• 30. **Mathematical Frontiers** and the future of mathematics.